

Sensor structure shapes the format of cognitive maps in navigation agents

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I. TWO ROUTES TO SPATIAL INFORMATION, ONE FINITE MEMORY

A PointGoal navigation (PointNav) agent is a deep-RL system that, given a goal coordinate, must walk a 3D scene to reach it. Trained agents develop *cognitive-map-like* internal representations of space inside their recurrent (LSTM) memory — documented even in agents *without* any visual input [?]. The agent has two routes to the spatial information its policy needs to act each step: (i) recompute it from the current visual encoder output, or (ii) maintain it across time in the recurrent memory by integrating the GPS sensor signal. These routes compete for finite hidden-state capacity: does visual input structure determine which route wins, and therefore the *format* the cognitive map takes?

The capacity-allocation account answers *yes*. A bottleneck at the encoder leaves only the integration route, so the LSTM is forced to carry a temporally-stable, linearly-readable position code in the policy-readable layer ($\mathbf{h}_2$) — the cognitive map. A rich encoder opens the visual-feature route; policy gradient reallocates capacity to it, and the linearly-readable component of $\mathbf{h}_2$ collapses — though the same hidden state still encodes position, only non-linearly. *Same task, different cognitive-map format.*

II. SETUP AND METHOD

We hold architecture, training, and the non-visual sensor stack constant across five conditions and vary only the visual input (Table I). All agents share a 3-layer LSTM (512-d) and the standard goal-in-start-frame / GPS / compass / close-to-goal sensor stack of [?]. We retrained all four sighted conditions from scratch under unified hyperparameters (250M frames each) to fix the inconsistency in our *Report 1* runs. All achieve 96–99% task success and $\text{SPL} \in [0.85, 0.89]$, ruling out a skill-gap explanation for any cross-condition representational difference.

Table I
 FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT (SIZE OF THE VISUAL FEATURE MAP PER TIMESTEP). BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred (σ_{max})
Uniform	4×4	256×256 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes and run six probes on the top-layer hidden state $\mathbf{h}_2$. A *probe* is a small regression model trained to decode some target (e.g. the agent’s GPS) from $\mathbf{h}_2$ with 5-fold

episode-level cross-validation; high R^2 means the target is decodable. We use both *linear* probes (Ridge $\alpha=10$ — this measures what a linear policy head, the operation DD-PPO actually performs, could read) and 2-layer *MLP* probes (this measures what is encoded at all, linear-readable or not). We also run lag- k probes (decoding past position GPS_{t-k} from current $\mathbf{h}_t$), per-(unit, scene) Skaggs spatial information from the place-cell literature, cross-training probes at 5 checkpoints across training, and cross-condition CKA / Procrustes shape distance / memory transplant.

III. ENCODER BANDWIDTH SHAPES THE LINEAR AXIS

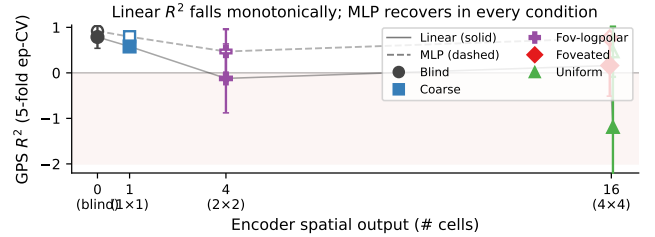


Figure 1. Top-layer GPS R^2 (mean $\pm \sigma$) vs. encoder spatial output. Solid: linear (Ridge) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

The cleanest summary of the post-retrain data is Figure 1 (solid line): linear GPS R^2 drops smoothly as encoder spatial output grows. The trend gave us a falsification target: foveated-logpolar (a 2×2 -output rich-encoder control we added *after* Report 1) was predicted to land in the rich-encoder regime, and Figure 1 shows it does — ruling out the alternative that foveation *specifically* (rather than encoder bandwidth in general) is what shapes the cognitive map. The other obvious alternative — that the linear probe simply lacks capacity in rich-encoder hidden states — is ruled out by distance-to-goal, a same-dimensional control target every condition decodes at $R^2 > 0.8$.

Yet position information is not destroyed by a rich encoder. A non-linear MLP probe on the *same* hidden states recovers GPS in every condition (Figure 1, dashed), and the gap from linear to MLP widens with encoder bandwidth. The combined picture is the capacity-allocation account in operation: the bottleneck (no encoder, or 1×1) leaves only the integration route, so the LSTM stores position in a linearly-readable axis; a rich encoder opens a per-step visual-feature route, policy gradient reallocates capacity to it, and the integrated code stops being the linearly-readable axis — though, the MLP recovery shows, position remains encoded. Format differs; total information shifts only modestly.

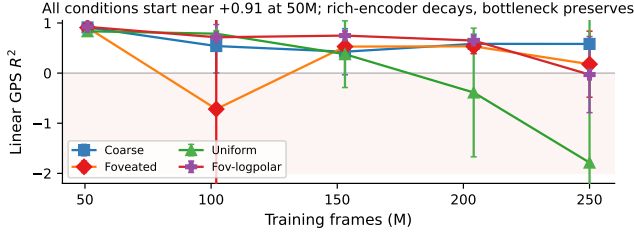


Figure 2. Linear GPS R^2 across training (5 checkpoints per condition; blind has only the 340M ckpt and is omitted). Negative- R^2 outliers clipped to -2.0 .

IV. ALLOCATION EMERGES ACROSS TRAINING

The endpoint contrast leaves open whether bottleneck conditions ever learned the visual route. Re-running the linear probe at intermediate checkpoints settles this: all four sighted conditions begin training with a high linear top-layer GPS code (Figure 2); the linear axis is the LSTM’s *default* format. What separates conditions is what happens *after* — bottleneck preserves the integrated code; the three rich-encoder conditions decay as the visual route consolidates, with decay rate ordered by encoder informativeness about position. Capacity allocation is therefore a training-time process, not an endpoint property; bottleneck conditions did not avoid the visual route by accident, they kept the integration route on top because no usable visual alternative emerged.

V. BOTTLENECK INTEGRATES; RICH ENCODER REACTS

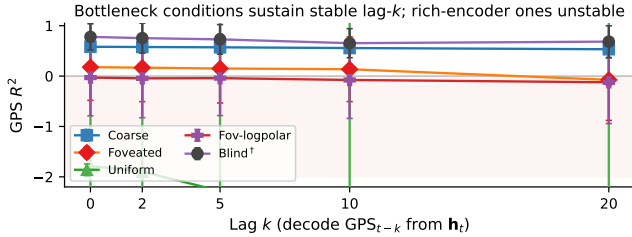


Figure 3. GPS R^2 for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Blind point: preliminary 100-ep partial NPZ.

The same format difference appears within an episode. Decoding past position from current $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustain stable positive R^2 across the lag range while rich-encoder ones degrade rapidly: a smoothly-varying *integrated* code preserves past-position information; a *reactive* code keyed to current visual context does not. As a final defence against the obvious null — “maybe bottleneck conditions just have more place cells” — per-(unit, scene) Skaggs spatial information is essentially condition-flat at $[1.16, 1.35]$ bits, well within the within-condition spread. The bandwidth-allocation signal lives in the linear-readout axis, not the per-unit place-cell metric.

VI. DEVIATIONS FROM THE PROPOSAL

H1 reframed. The proposal cast H1 as compensatory memory specifically in the foveated agent, and *Report 1*

(on partially-converged checkpoints with heterogeneous hyperparameters) reported foveated as the longest-memory condition. Both are now superseded. After unified retraining — which removed the apparent foveated specialness — and after adding foveated-logpolar as a falsification test, the data show a clean encoder-bandwidth trend (Figure 1) under which foveated and uniform sit in the same rich-encoder regime. *The mechanism is encoder bandwidth, not foveation specifically.*

H3 de-scoped, methodology broadened. The proposal’s H3 (epistemic gaze: a learned-gaze MLP targeting memory-weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward (a bounded- σ stochastic-gaze fix is in the codebase but not pursued this iteration). Compensating, we imported a richer set of cognitive-neuroscience methods into the representational-analysis frame — Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, an excursion-forgetting protocol, and a leave-one-scene-out scene-invariance test. *The contribution shifts from a single epistemic-gaze deliverable to recasting classical cognitive-neuroscience metrics as a computational framework for representational analysis of deep-RL agents.*

VII. OPEN ISSUES AND ACTION ITEMS

Encountered and solved. (i) Hyperparameter drift across Report 1 conditions confounded comparisons; resolved by unified retraining. (ii) Skaggs spatial information $\sim 100\times$ inflated on raw $\mathbf{h}_2$ (near-zero global means dominate the ratio in $I = \sum p(b)(\lambda_b/\bar{\lambda}) \log_2(\lambda_b/\bar{\lambda})$); resolved by ReLU rectification ($\max(\mathbf{h}_2, 0)$) before computing per-bin firing rates. (iii) Coarse memory transplant failed at state-dict load (RGB shape mismatch); restricted to the 6 sighted-only pairs. (iv) Blind probe-collect at 50 episodes gave wide CV folds (-0.27 ± 1.41); diagnosed as sample-size noise (100 eps stabilises to $+0.79 \pm 0.25$).

Forward risks. Blind 500-ep ETA overlaps the NeurIPS deadline; single-seed per condition leaves seed-variance unbounded; the transplant matrix is incomplete without coarse / blind pairs, deferring the H2 format-axis behavioural test until an input-padding fix lands.

Toward the final report — bandwidth-allocation claim. (A1) complete blind 500-ep probe-collect, populate the blind row across all lenses, tighten Figures 1 and 3; (A2) retrain coarse with a 256×256 -padded input pipeline so the transplant matrix can include the bottleneck-vs-rich-encoder asymmetry test; (A3) two-seed validation on the coarse and uniform endpoints. *Cog-neuro methodology.* (A4) report the LOSO scene-invariance and excursion-forgetting results in the final write-up; (A5) place-vs-view-cell dissociation on existing rate-map data, plus an H3 reattempt with bounded- σ stochastic gaze if time permits.